```
CREATE TABLE student_study_habits (
    student_id VARCHAR(10) PRIMARY KEY,
    age INT,
    gender VARCHAR(20),
    city VARCHAR(50),
    state VARCHAR(50),
    course VARCHAR(50),
    branch VARCHAR(50),
    year_of_study VARCHAR(20),

    family_income NUMERIC(10,2),
    parents_education VARCHAR(50),

    internet_access VARCHAR(10),
    device_type VARCHAR(30),

    study_hours_per_day NUMERIC(4,2),
    self_study_hours NUMERIC(4,2),
    online_class_hours NUMERIC(4,2),
    coaching_hours NUMERIC(4,2),

    sleep_hours_per_day NUMERIC(4,2),
    exercise_hours_per_week NUMERIC(4,2),

    social_media_hours NUMERIC(4,2),
    gaming_hours NUMERIC(4,2),
```

```
ott_hours NUMERIC(4,2),

 attendance_percentage NUMERIC(5,2),
 assignment_submission_rate NUMERIC(5,2),

 projects_completed INT,
 seminar_attendance INT,
 library_visits_per_month INT,
 doubt_clearing_sessions INT,

 exam_preparation_days INT,

 previous_semester_percentage NUMERIC(5,2),
 current_semester_percentage NUMERIC(5,2),

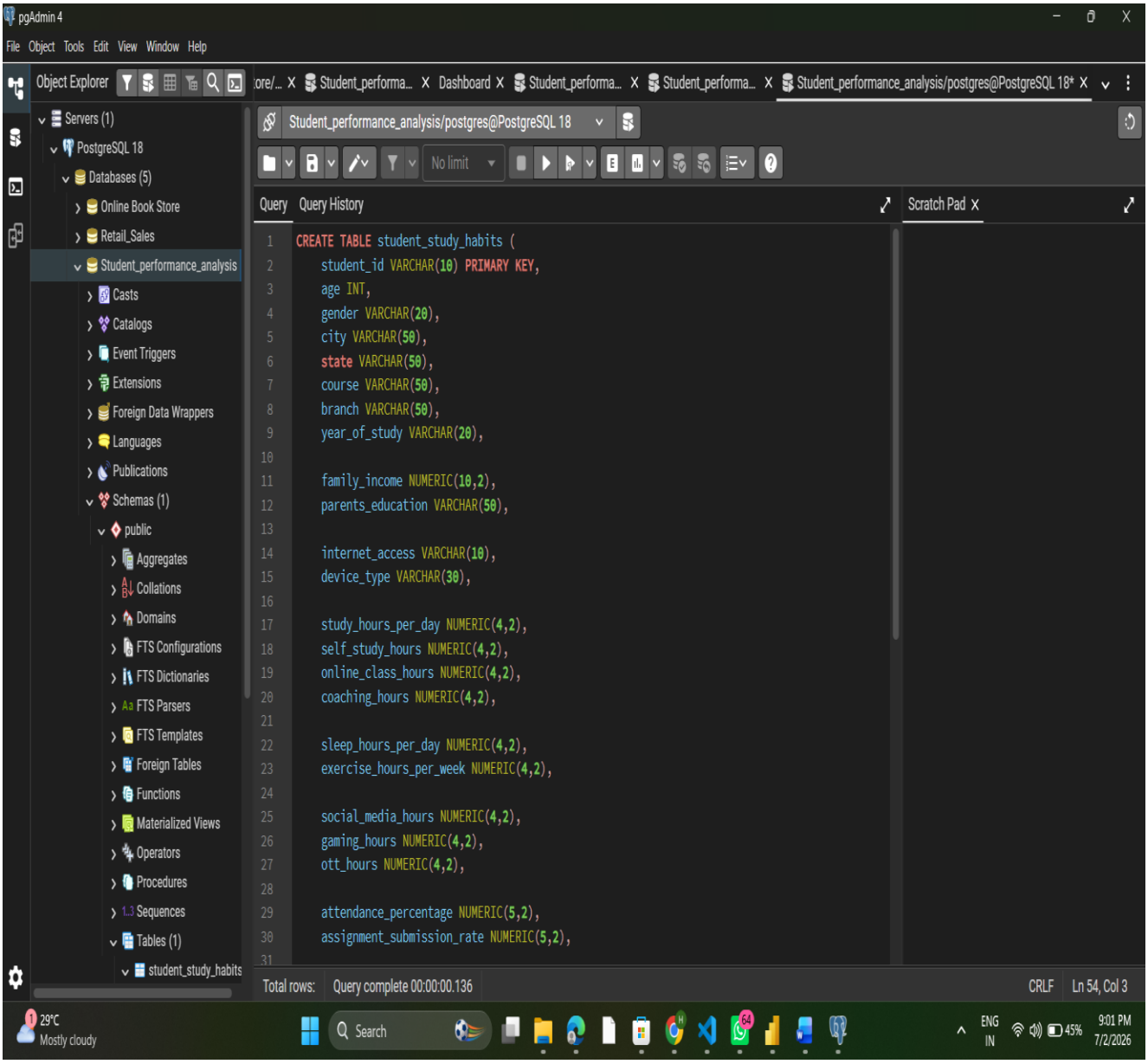
 internal_assessment_score NUMERIC(5,2),
 mock_test_score NUMERIC(5,2),

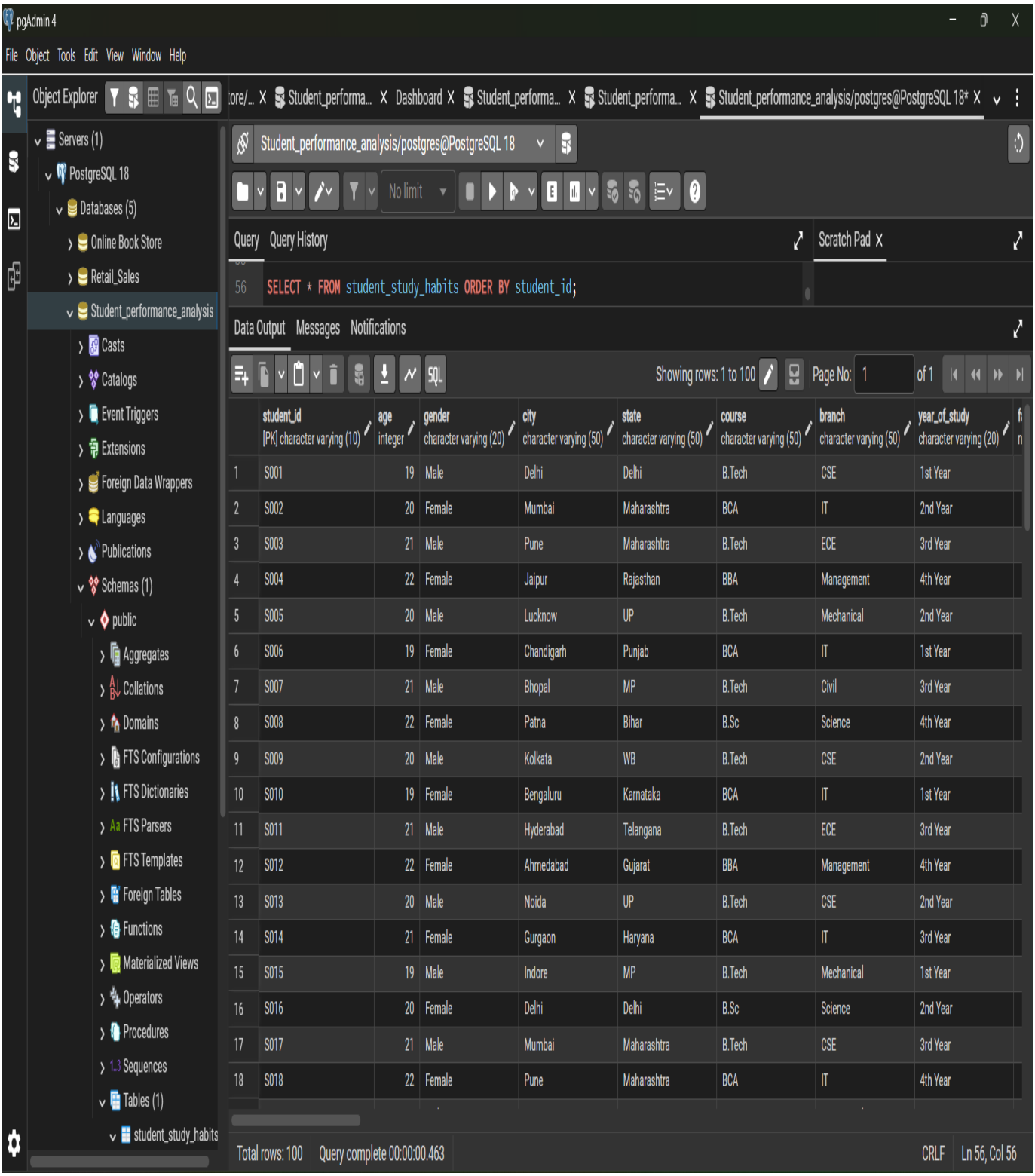
 stress_level INT,
 motivation_level INT,
 time_management_score INT,
 health_rating INT,

 part_time_job VARCHAR(10),
 scholarship_status VARCHAR(10),
```

final_performance_category VARCHAR(20)

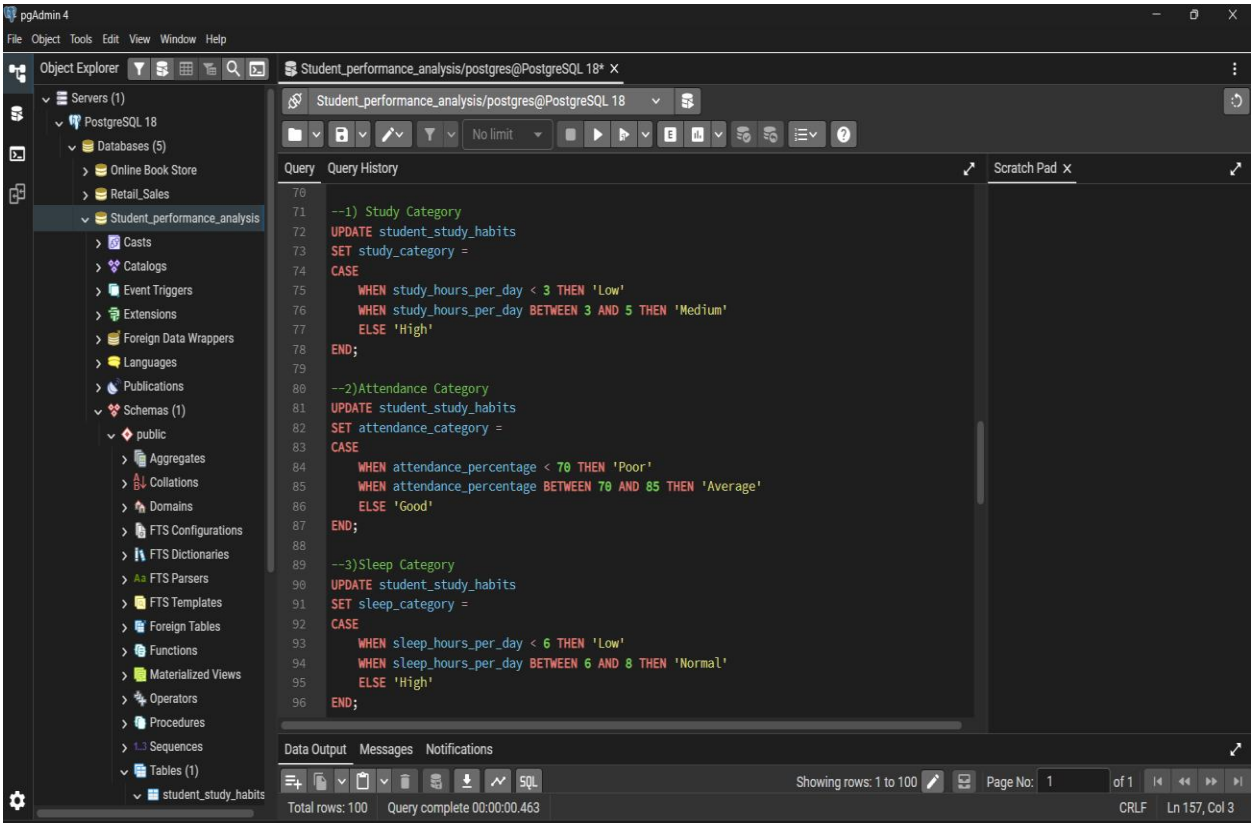
);

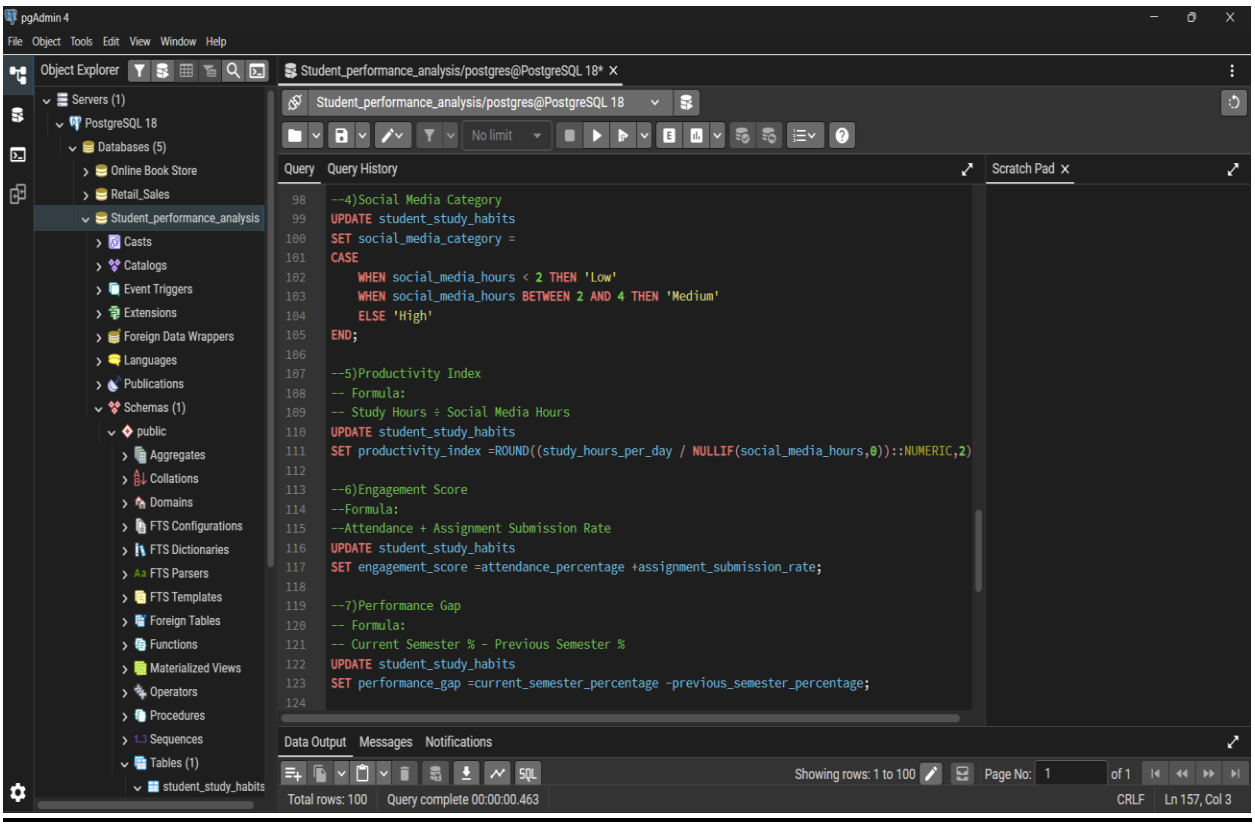


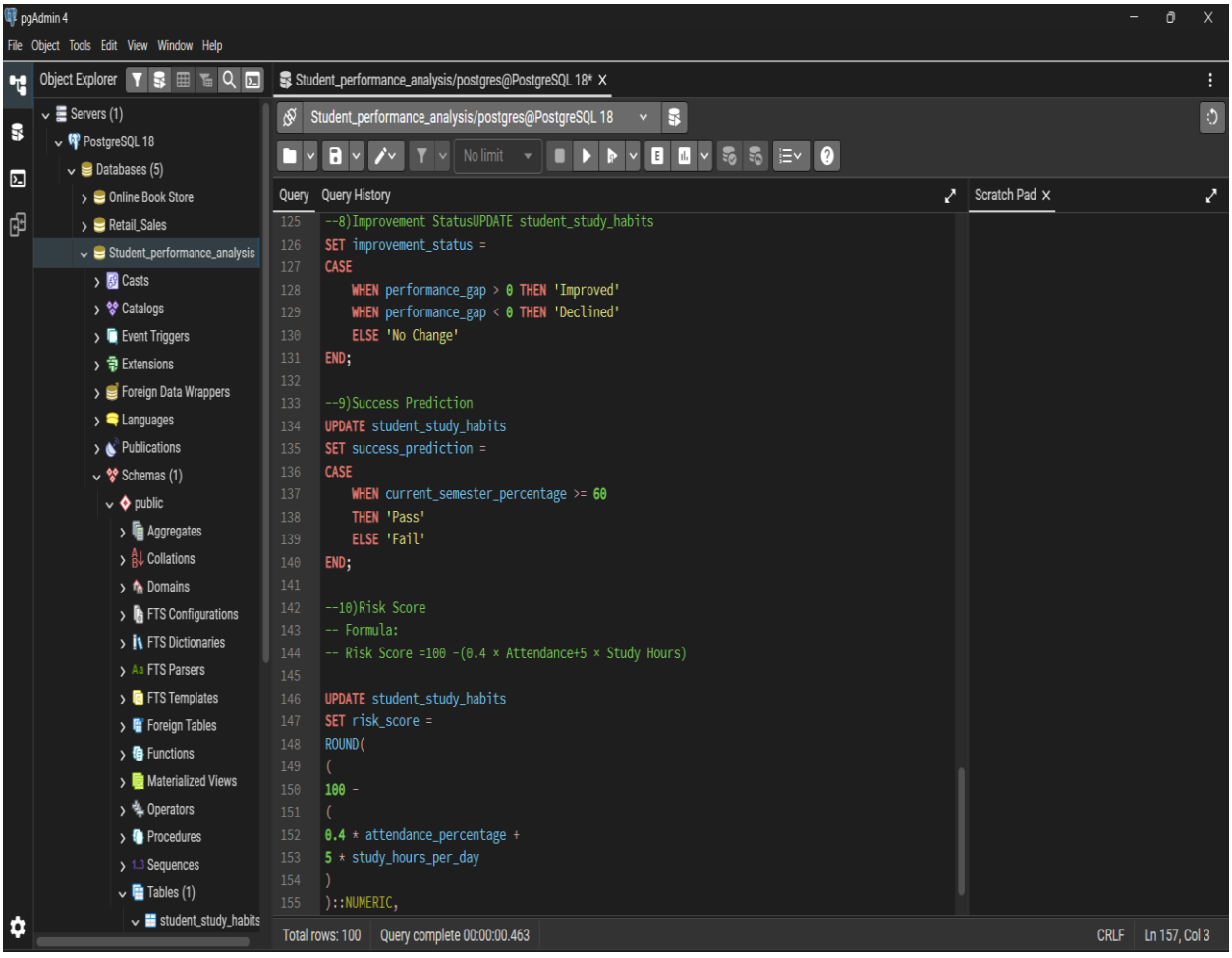

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56 SELECT * FROM student_study_habits ORDER BY student_id;
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Showing rows: 1 to 100
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of 1
student_id [PK] character varying (10)
age integer
gender character varying (20)
city character varying (50)
state character varying (50)
course character varying (50)
branch character varying (50)
year_of_study character varying (20)
1 S001 19 Male Delhi Delhi B.Tech CSE 1st Year
2 S002 20 Female Mumbai Maharashtra BCA IT 2nd Year
3 S003 21 Male Pune Maharashtra B.Tech ECE 3rd Year
4 S004 22 Female Jaipur Rajasthan BBA Management 4th Year
5 S005 20 Male Lucknow UP B.Tech Mechanical 2nd Year
6 S006 19 Female Chandigarh Punjab BCA IT 1st Year
7 S007 21 Male Bhopal MP B.Tech Civil 3rd Year
8 S008 22 Female Patna Bihar B.Sc Science 4th Year
9 S009 20 Male Kolkata WB B.Tech CSE 2nd Year
10 S010 19 Female Bengaluru Karnataka BCA IT 1st Year
11 S011 21 Male Hyderabad Telangana B.Tech ECE 3rd Year
12 S012 22 Female Ahmedabad Gujarat BBA Management 4th Year
13 S013 20 Male Noida UP B.Tech CSE 2nd Year
14 S014 21 Female Gurgaon Haryana BCA IT 3rd Year
15 S015 19 Male Indore MP B.Tech Mechanical 1st Year
16 S016 20 Female Delhi Delhi B.Sc Science 2nd Year
17 S017 21 Male Mumbai Maharashtra B.Tech CSE 3rd Year
18 S018 22 Female Pune Maharashtra BCA IT 4th Year
Total rows: 100
Query complete 00:00:00.463
CRLF
Ln 56, Col 56

Add Derived Columns

```
ALTER TABLE student_study_habits
ADD COLUMN study_category VARCHAR(20),
ADD COLUMN attendance_category VARCHAR(20),
ADD COLUMN sleep_category VARCHAR(20),
ADD COLUMN social_media_category VARCHAR(20),
ADD COLUMN risk_score NUMERIC(10,2),
ADD COLUMN productivity_index NUMERIC(10,2),
ADD COLUMN engagement_score NUMERIC(10,2),
ADD COLUMN performance_gap NUMERIC(10,2),
ADD COLUMN success_prediction VARCHAR(20),
ADD COLUMN improvement_status VARCHAR(20);
```


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Student_performance_analysis/postgres@PostgreSQL 18
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Query
Query History
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70
71 --1) Study Category
72 UPDATE student_study_habits
73 SET study_category =
74 CASE
75 WHEN study_hours_per_day < 3 THEN 'Low'
76 WHEN study_hours_per_day BETWEEN 3 AND 5 THEN 'Medium'
77 ELSE 'High'
78 END;
79
80 --2)Attendance Category
81 UPDATE student_study_habits
82 SET attendance_category =
83 CASE
84 WHEN attendance_percentage < 70 THEN 'Poor'
85 WHEN attendance_percentage BETWEEN 70 AND 85 THEN 'Average'
86 ELSE 'Good'
87 END;
88
89 --3)Sleep Category
90 UPDATE student_study_habits
91 SET sleep_category =
92 CASE
93 WHEN sleep_hours_per_day < 6 THEN 'Low'
94 WHEN sleep_hours_per_day BETWEEN 6 AND 8 THEN 'Normal'
95 ELSE 'High'
96 END;
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Total rows: 100
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Query
Query History
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98 --4)Social Media Category
99 UPDATE student_study_habits
100 SET social_media_category =
101 CASE
102 WHEN social_media_hours < 2 THEN 'Low'
103 WHEN social_media_hours BETWEEN 2 AND 4 THEN 'Medium'
104 ELSE 'High'
105 END;
106
107 --5)Productivity Index
108 -- Formula:
109 -- Study Hours ÷ Social Media Hours
110 UPDATE student_study_habits
111 SET productivity_index =ROUND((study_hours_per_day / NULLIF(social_media_hours,0))::NUMERIC,2)
112
113 --6)Engagement Score
114 --Formula:
115 --Attendance + Assignment Submission Rate
116 UPDATE student_study_habits
117 SET engagement_score =attendance_percentage +assignment_submission_rate;
118
119 --7)Performance Gap
120 -- Formula:
121 -- Current Semester % - Previous Semester %
122 UPDATE student_study_habits
123 SET performance_gap =current_semester_percentage -previous_semester_percentage;
124
Data Output
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Total rows: 100
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Student_performance_analysis/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad X
--8)Improvement StatusUPDATE student_study_habits
SET improvement_status =
CASE
WHEN performance_gap > 0 THEN 'Improved'
WHEN performance_gap < 0 THEN 'Declined'
ELSE 'No Change'
END;
--9)Success Prediction
UPDATE student_study_habits
SET success_prediction =
CASE
WHEN current_semester_percentage >= 60
THEN 'Pass'
ELSE 'Fail'
END;
--10)Risk Score
-- Formula:
-- Risk Score =100 -(0.4 × Attendance+5 × Study Hours)
UPDATE student_study_habits
SET risk_score =
ROUND(
(
100 -
(
0.4 * attendance_percentage +
5 * study_hours_per_day
)
)::NUMERIC,
Total rows: 100
Query complete 00:00:00.463
CRLF
Ln 157, Col 3